GRACE: Gradient-Free Robot Action Generation via Combined Diffusion–MPPI Posterior Estimation

***Abstract*—Diffusion policies generate multimodal robot action sequences from demonstrations, but steering them toward deployment-time constraints typically relies on differentiable guidance costs. This excludes many practical safety constraints, such as binary collision checks, joint limits, and black-box rollout costs, that are nondifferentiable. We propose Gradient-free Robot Action generation via Combined diffusion–MPPI posterior Estimation (GRACE), which guides a pretrained diffusion policy with Model Predictive Path Integral (MPPI) control using only forward cost evaluations. We show that the diffusion reverse step and the MPPI update share a common score-ascent structure. This lets us recast each constrained reverse step as a Bayesian posterior and approximate its mean with a single MPPI update centered at the diffusion prior mean, integrated directly into denoising rather than wrapped around it as an outer loop. Under this view, gradient guidance emerges as a special case that holds only when the cost is differentiable. GRACE thus enables constraint-aware generation without retraining the diffusion model or designing differentiable surrogates. Across simulation and real-world robot experiments, GRACE attains higher success rates and fewer safety violations than both gradient-guided diffusion policies and sampling-based planners. Code and experiment videos are available at https://anonymous.4open.science/w/grace-70BB/.**

***Index Terms*—Diffusion models, model predictive path integral control, motion planning, posterior sampling, gradient-free optimization.**

I. INTRODUCTION

ROBOT trajectory generation is a longstanding problem in robotics. An autonomous system must produce trajectories that are dynamically feasible, satisfy task objectives, and remain adaptable to environments that change between deployments. Traditional planners such as A* [1], Probabilistic Roadmap (PRM) [2], and Rapidly-exploring Random Tree (RRT) [3] explore the state or configuration space to find feasible paths, while trajectory optimizers such as Covariant Hamiltonian Optimization for Motion Planning (CHOMP) [4], Stochastic Trajectory Optimization for Motion Planning (STOMP) [5], and Gaussian Process Motion Planning (GPMP) [6] refine an initial trajectory by minimizing costs that encode feasibility, smoothness, and task objectives. These methods have served as the workhorse of deployed robotic systems for decades, but they typically converge to a single mode of the solution space and cannot capture the multiple qualitatively distinct trajectories that are valid in many manipulation and navigation tasks [7].

Learning-based methods address this by modeling the distribution of valid trajectories from demonstration data. In particular, diffusion generative models [8], [9] represent high-dimensional and multimodal distributions by iteratively denoising Gaussian noise, and have been applied to robot trajectory generation [10] and visuomotor policy learning [11]. Such a prior captures the multimodal structure of the demonstrations, but it remains a model that samples from the learned distribution and, on its own, does not specify how those samples should be steered toward a particular task objective or constraint. Providing this task-level steering is the role of guidance.

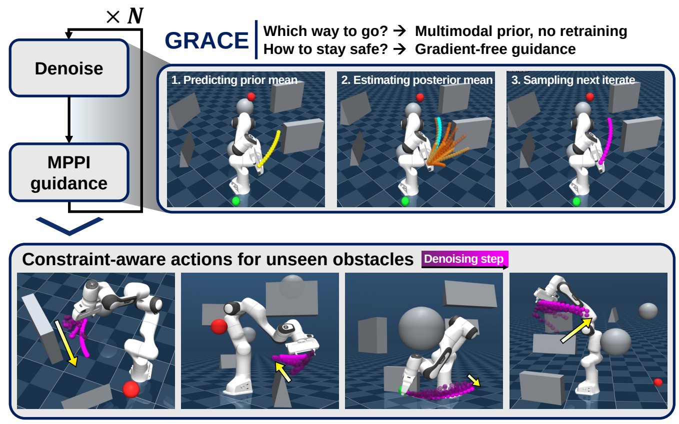


Fig. 1. The multimodal diffusion prior decides which way to go without retraining, while gradient-free guidance decides how to stay safe. At each denoising step (top), GRACE predicts the diffusion prior mean (yellow), estimates an MPPI posterior mean (cyan) from gradient-free rollouts (orange), and samples the next iterate (magenta). Over the denoising process (bottom), the trajectory is refined into a constraint-aware one that avoids unseen obstacles. The green and red markers denote the start and goal.

Guidance can inject the task objective either during training or at inference time. Classifier [12] and classifier-free guidance [13] fold the objective into the prior at training time, but require the relevant constraints to be known in advance. Deployment-time constraints, however, such as obstacles introduced only after training, are unavailable when the prior is learned, so the prior must instead be steered at inference time without retraining [14]. Cost-gradient guidance adds the gradient of a task cost to each reverse step [15], [16], barrier-based methods embed safety specifications through control barrier functions [17], and related schemes use the gradient of a learned dynamics model [18]. These approaches require the objective to be differentiable, which many practical constraints, such as binary collision checks, joint limits, and black-box rollout costs, are not, and they remain prone to local minima when the cost landscape is non-convex [19]. Diffusion-ES [20] avoids differentiability by scoring candidates through forward rollouts of the cost, but attaches the sampling optimizer as an outer loop around the diffusion model, leaving the prior and the guidance as two separate components rather than a single guided generation step.

We instead integrate sampling-based guidance directly into the reverse-diffusion process, using Model Predictive Path Integral (MPPI) [21] control. As illustrated in Fig. 1, each reverse step first denoises with the diffusion prior and then applies an MPPI update that scores sampled action candidates

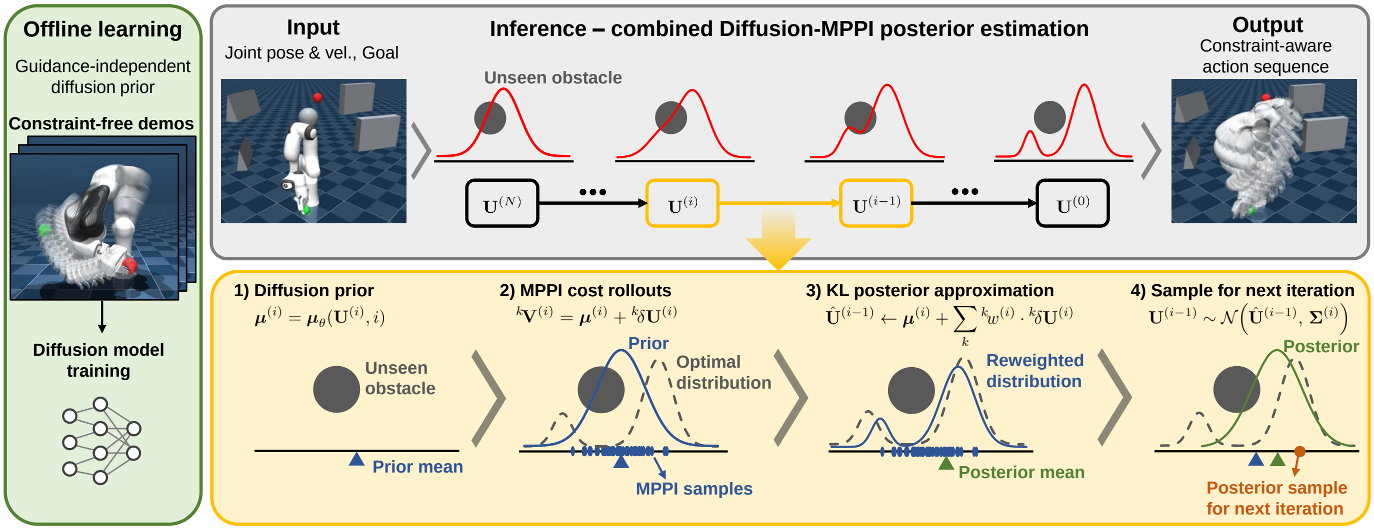


Fig. 2. The GRACE inference pipeline. A pretrained diffusion prior, learned offline from constraint-free demonstrations, proposes an action-sequence distribution, and at each reverse step an MPPI update reweights the sampled candidates using deployment-time constraint costs. The resulting posterior approximation steers the reverse diffusion toward constraint-aware action sequences without differentiating the cost or the dynamics.

through forward rollouts of the cost, so the framework is gradient-free, requires no retraining, and handles nondifferentiable costs such as binary collision checks and joint limits. Crucially, this update is applied within a single reverse step rather than as an outer loop around the diffusion model, so the prior and the guidance form one guided generation step. This integration follows from a structural similarity between the two: each can be written as a score-ascent step that moves a sample toward higher density of an underlying distribution, learned offline for diffusion and estimated online for MPPI. Building on this, we recast each constrained reverse step as a Bayesian posterior whose tractable approximation is an MPPI-style update centered at the diffusion prior mean. When the cost is differentiable, gradient guidance emerges as a special case of the same approximation, making the framework a more general view of inference-time guidance.

The contributions of this paper are as follows:

- **Unifying diffusion and MPPI for guidance.** The score-ascent structure shared by the two updates is exploited to integrate the sampling-based guidance of MPPI directly into the diffusion reverse step, rather than wrapping it around the model as an outer loop.
- **Constraint-aware generation without gradients.** Each constrained reverse step is formulated as a Bayesian posterior whose Gaussian approximation is an MPPI update centered at the diffusion prior mean, enabling guidance under nondifferentiable costs without differentiable surrogates or retraining.
- **Improved success and safety.** GRACE is validated in simulation and on real hardware, attaining higher success rates and fewer safety violations than gradient-guided diffusion and sampling-based baselines while preserving the multimodality of the prior.

II. Preliminaries

A. Diffusion Models for Trajectory Generation

Let $\mathbf{U}^{(0)} \in \mathbb{R}^{H\times m}$ denote an action sequence of horizon H drawn from a demonstration distribution p_{data}. The forward diffusion process [22] gradually corrupts $\mathbf{U}^{(0)}$ by Gaussian noise over N steps,

$$q(\mathbf{U}^{(i)} \mid \mathbf{U}^{(i-1)}) = \mathcal{N}\Big(\mathbf{U}^{(i)}; \sqrt{\alpha^{(i)}}\,\mathbf{U}^{(i-1)},\, (1-\alpha^{(i)})\mathbf{I}\Big), \quad (1)$$

where $\alpha^{(i)} = 1 - \beta^{(i)}$ for a predefined noise schedule $\{\beta^{(i)}\}_{i=1}^N$. The i-step forward noising distribution can be written in closed form as

$$q(\mathbf{U}^{(i)} \mid \mathbf{U}^{(0)}) = \mathcal{N}\Big(\mathbf{U}^{(i)}; \sqrt{\bar{\alpha}^{(i)}}\,\mathbf{U}^{(0)},\, (1-\bar{\alpha}^{(i)})\mathbf{I}\Big), \quad (2)$$

where $\bar{\alpha}^{(i)} = \prod_{j=1}^{i} \alpha^{(j)}$. Equivalently, it can be reparameterized as

$$\mathbf{U}^{(i)} = \sqrt{\bar{\alpha}^{(i)}}\,\mathbf{U}^{(0)} + \sqrt{1-\bar{\alpha}^{(i)}}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}). \quad (3)$$

Generation reverses this process through a learned Gaussian kernel

$$p_\theta(\mathbf{U}^{(i-1)} \mid \mathbf{U}^{(i)}) = \mathcal{N}\Big(\mathbf{U}^{(i-1)};\, \boldsymbol{\mu}_\theta(\mathbf{U}^{(i)}, i),\, \boldsymbol{\Sigma}^{(i)}\Big), \quad (4)$$

whose mean is parameterized through a learned noise predictor $\boldsymbol{\epsilon}_\theta$ as

$$\boldsymbol{\mu}_\theta(\mathbf{U}^{(i)}, i) = \frac{1}{\sqrt{\alpha^{(i)}}}\left(\mathbf{U}^{(i)} - \frac{\beta^{(i)}}{\sqrt{1-\bar{\alpha}^{(i)}}}\,\boldsymbol{\epsilon}_\theta(\mathbf{U}^{(i)}, i)\right), \quad (5)$$

trained by minimizing the denoising objective

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{i,\mathbf{U}^{(0)},\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{U}^{(i)}, i)\right\|_2^2\right]. \quad (6)$$

B. Model Predictive Path Integral Control

Consider a discrete-time system $\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{v}_t)$ with state $\mathbf{x}_t \in \mathbb{R}^n$ and applied control $\mathbf{v}_t \in \mathbb{R}^m$. Let $\mathbf{V} =$

$(\mathbf{v}_0, \ldots, \mathbf{v}_{H-1})$ denote an applied control sequence. Given $\mathbf{x}_0$, the rollout cost is

$$J(\mathbf{V}; \mathbf{x}_0) = \sum_{t=0}^{H-1} \ell(\mathbf{x}_t, \mathbf{v}_t) + \phi(\mathbf{x}_H). \tag{7}$$

The information-theoretic formulation [23] yields the optimal control distribution

$$q^\star(\mathbf{V}) = \frac{1}{\eta} \exp\big(- J(\mathbf{V}; \mathbf{x}_0)/\lambda\big)\, \mathcal{N}(\mathbf{V}; \widetilde{\mathbf{U}}, \mathbf{\Sigma}), \tag{8}$$

where $\lambda > 0$ is a temperature, $\mathcal{N}(\mathbf{V}; \widetilde{\mathbf{U}}, \mathbf{\Sigma})$ is a Gaussian base measure centered at $\widetilde{\mathbf{U}}$, and the normalizer is

$$\eta = \int \exp\big(- J(\mathbf{V}; \mathbf{x}_0)/\lambda\big)\, \mathcal{N}(\mathbf{V}; \widetilde{\mathbf{U}}, \mathbf{\Sigma})\, d\mathbf{V}. \tag{9}$$

The common uncontrolled choice is $\widetilde{\mathbf{U}} = 0$. MPPI approximates $q^\star$ within the Gaussian family $\mathcal{N}(\mathbf{V}; \mathbf{U}, \mathbf{\Sigma})$, optimizing only the mean $\mathbf{U}$ while holding the covariance $\mathbf{\Sigma}$ fixed, by minimizing

$$\mathbf{U}^\star = \arg\min_{\mathbf{U}} D_{\mathrm{KL}}(q^\star(\mathbf{V}) \,\|\, \mathcal{N}(\mathbf{V}; \mathbf{U}, \mathbf{\Sigma}))\,. \tag{10}$$

For fixed $\mathbf{\Sigma}$, this gives the moment-matching solution

$$\mathbf{U}^\star = \mathbb{E}_{q^\star}[\mathbf{V}]. \tag{11}$$

Substituting (9) for the normalizer expresses this moment as a ratio of expectations over the proposal $\mathcal{N}(\mathbf{V}; \mathbf{U}, \mathbf{\Sigma})$. Because the base measure and the proposal share the covariance $\mathbf{\Sigma}$, the Gaussian ratio between them absorbs into the cost, giving

$$\mathbf{U}^\star = \frac{\mathbb{E}_{\mathcal{N}(\mathbf{U},\mathbf{\Sigma})}\big[\mathbf{V}\, \exp(-\widetilde{J}(\mathbf{V})/\lambda)\big]}{\mathbb{E}_{\mathcal{N}(\mathbf{U},\mathbf{\Sigma})}\big[\exp(-\widetilde{J}(\mathbf{V})/\lambda)\big]}, \tag{12}$$

where the augmented cost is

$$\widetilde{J}(\mathbf{V}; \mathbf{U}, \widetilde{\mathbf{U}}, \mathbf{x}_0) = J(\mathbf{V}; \mathbf{x}_0) + \lambda \sum_{t=0}^{H-1} (\mathbf{u}_t - \widetilde{\mathbf{u}}_t)^\top \mathbf{\Sigma}^{-1} \mathbf{v}_t. \tag{13}$$

Using samples ${}^k\mathbf{V} = \mathbf{U} + {}^k\delta\mathbf{U}$ with ${}^k\delta\mathbf{U} \sim \mathcal{N}(0, \mathbf{\Sigma})$, the Monte Carlo estimator of (12) is

$$\mathbf{U}^\star = \mathbf{U} + \sum_{k=1}^{K} {}^k w \cdot {}^k\delta\mathbf{U}, \tag{14}$$

where

$${}^k w = \frac{\exp\big(- \widetilde{J}({}^k\mathbf{V})/\lambda\big)}{\sum_{l=1}^{K} \exp\big(- \widetilde{J}({}^l\mathbf{V})/\lambda\big)}. \tag{15}$$

III. PROPOSED METHOD

In this section, we present GRACE, a framework that guides a pretrained diffusion policy with MPPI by casting each constrained reverse step as a Bayesian posterior. The starting point is a structural observation that the diffusion reverse step and the MPPI update share a common score-ascent structure, with each advancing its iterate along the scaled score of an underlying distribution. We develop this observation into our framework over the course of the section.

A. Diffusion and MPPI as Score Ascent

We make the score-ascent correspondence explicit, as the combination in Section III-B builds on it. Differentiating the log of (2) yields

$$\nabla_{\mathbf{U}^{(i)}} \log q(\mathbf{U}^{(i)} \mid \mathbf{U}^{(0)}) = \frac{-\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}^{(i)}}}, \tag{16}$$

so the noise target $\boldsymbol{\epsilon}$ is proportional to the score of the forward marginal. At the optimum of the denoising objective (6), the noise predictor recovers the conditional expectation of $\boldsymbol{\epsilon}$, so that the learned score

$$\boldsymbol{s}_\theta(\mathbf{U}^{(i)}, i) = -\frac{\boldsymbol{\epsilon}^\star_\theta(\mathbf{U}^{(i)}, i)}{\sqrt{1-\bar{\alpha}^{(i)}}} \tag{17}$$

matches the score of the noise-perturbed marginal $q^{(i)}$, i.e. $\boldsymbol{s}_\theta(\mathbf{U}^{(i)}, i) = \nabla_{\mathbf{U}^{(i)}} \log q^{(i)}(\mathbf{U}^{(i)})$. Substituting (17) into (5) and neglecting the prefactor $1/\sqrt{\alpha^{(i)}}$ expresses the reverse mean as a score ascent step,

$$\boldsymbol{\mu}_\theta(\mathbf{U}^{(i)}, i) \approx \mathbf{U}^{(i)} + \beta^{(i)}\, \boldsymbol{s}_\theta(\mathbf{U}^{(i)}, i). \tag{18}$$

The same view extends to MPPI. Dropping the control prior in (13) reduces the augmented cost to J, so the denominator of (12) becomes

$$\Psi(\mathbf{U}) := \mathbb{E}_{\mathbf{V}\sim\mathcal{N}(\mathbf{U},\mathbf{\Sigma})}\big[e^{-J(\mathbf{V})/\lambda}\big], \tag{19}$$

and normalizing it defines $\pi(\mathbf{U}) := \Psi(\mathbf{U})/Z$. Substituting $\mathbf{V} = \mathbf{U} + (\mathbf{V} - \mathbf{U})$ into (12) gives

$$\mathbf{U}^\star = \mathbf{U} + \frac{\mathbb{E}_{\mathcal{N}(\mathbf{U},\mathbf{\Sigma})}\big[(\mathbf{V}-\mathbf{U})\, e^{-J(\mathbf{V})/\lambda}\big]}{\Psi(\mathbf{U})}. \tag{20}$$

Using the Gaussian identity $\nabla_{\mathbf{U}} \mathcal{N}(\mathbf{V}; \mathbf{U}, \mathbf{\Sigma}) = \mathbf{\Sigma}^{-1}(\mathbf{V} - \mathbf{U})\, \mathcal{N}(\mathbf{V}; \mathbf{U}, \mathbf{\Sigma})$ and that $e^{-J/\lambda}$ is independent of $\mathbf{U}$,

$$\nabla_{\mathbf{U}} \Psi(\mathbf{U}) = \mathbf{\Sigma}^{-1}\, \mathbb{E}_{\mathcal{N}(\mathbf{U},\mathbf{\Sigma})}\big[(\mathbf{V}-\mathbf{U})\, e^{-J(\mathbf{V})/\lambda}\big]. \tag{21}$$

Since Z is constant, $\nabla_{\mathbf{U}} \log \pi = \nabla_{\mathbf{U}} \Psi / \Psi$. Left-multiplying (21) by $\mathbf{\Sigma}$ and dividing by Ψ shows that the second term of (20) equals $\mathbf{\Sigma}\, \nabla_{\mathbf{U}} \log \pi(\mathbf{U})$, so

$$\mathbf{U}^\star = \mathbf{U} + \mathbf{\Sigma}\, \nabla_{\mathbf{U}} \log \pi(\mathbf{U}). \tag{22}$$

Thus the MPPI update (22) and the diffusion reverse step (18) share the same form, each advancing the iterate along a scaled score of an underlying distribution, $q^{(i)}$ for diffusion and π for MPPI, learned offline from demonstrations for diffusion and evaluated online from cost rollouts for MPPI. This shared structure is what lets us combine the two within a single reverse step.

B. Constraint-Conditioned Reverse Posterior

To steer the reverse process toward constraint-satisfying trajectories, we treat constraint satisfaction as a conditioning event $\mathcal{C}$ and apply Bayes' rule to the reverse step,

$$p(\mathbf{U}^{(i-1)} \mid \mathbf{U}^{(i)}, \mathcal{C}) = \frac{p(\mathcal{C} \mid \mathbf{U}^{(i-1)}, \mathbf{U}^{(i)})\, p_\theta(\mathbf{U}^{(i-1)} \mid \mathbf{U}^{(i)})}{p(\mathcal{C} \mid \mathbf{U}^{(i)})}. \tag{23}$$

Constraint satisfaction is determined by the action sequence $\mathbf{U}^{(i-1)}$ and is conditionally independent of $\mathbf{U}^{(i)}$ given $\mathbf{U}^{(i-1)}$, so the likelihood simplifies to $p(\mathcal{C} \mid \mathbf{U}^{(i-1)})$. Since the denominator is independent of $\mathbf{U}^{(i-1)}$,

$$p(\mathbf{U}^{(i-1)}|\mathbf{U}^{(i)},\mathcal{C}) \propto p(\mathcal{C} \mid \mathbf{U}^{(i-1)})\, p_\theta(\mathbf{U}^{(i-1)} \mid \mathbf{U}^{(i)}). \quad (24)$$

We take the Boltzmann likelihood $p(\mathcal{C} \mid \mathbf{U}^{(i-1)}) \propto e^{-J(\mathbf{U}^{(i-1)})/\lambda}$, consistent with the cost-induced target of Section III-A, where $J(\mathbf{U}^{(i-1)})$ is the rollout cost of the candidate sequence. Substituting this and the Gaussian reverse kernel (4), and writing $\boldsymbol{\mu}^{(i)} := \boldsymbol{\mu}_\theta(\mathbf{U}^{(i)}, i)$,

$$p(\mathbf{U}^{(i-1)}|\mathbf{U}^{(i)},\mathcal{C}) = \frac{1}{Z_{\mathcal{C}}^{(i)}}\, e^{-J(\mathbf{U}^{(i-1)})/\lambda}\, \mathcal{N}(\mathbf{U}^{(i-1)};\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)}), \quad (25)$$

where the normalizer $Z_{\mathcal{C}}^{(i)}$ is

$$Z_{\mathcal{C}}^{(i)} = \int e^{-J(\mathbf{U}^{(i-1)})/\lambda}\, \mathcal{N}(\mathbf{U}^{(i-1)};\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)})\, d\mathbf{U}^{(i-1)}. \quad (26)$$

The posterior (25) has the same product form as the MPPI optimal distribution (8), a Boltzmann cost factor weighting a Gaussian, except that the Gaussian is centered at the diffusion prior mean $\boldsymbol{\mu}^{(i)}$ rather than at the base-measure mean $\widetilde{\mathbf{U}}$. Because J contains binary, piecewise-constant indicators, the Boltzmann factor is non-Gaussian, so $Z_{\mathcal{C}}^{(i)}$ has no closed form and (25) is intractable to sample from within each reverse step. The next subsection approximates it by a tractable Gaussian.

C. Gaussian Approximation via KL Minimization

For a given reverse step i, we approximate the constraint-conditioned posterior by a Gaussian whose covariance is kept equal to the diffusion reverse covariance $\boldsymbol{\Sigma}^{(i)}$, while only its mean is optimized:

$$\widehat{\mathbf{U}}^{(i-1)} = \arg\min_{\boldsymbol{\nu}} D_{\mathrm{KL}}\Big(p(\mathbf{U}^{(i-1)} | \mathbf{U}^{(i)},\mathcal{C}) \Big\| \mathcal{N}(\mathbf{U}^{(i-1)};\boldsymbol{\nu},\boldsymbol{\Sigma}^{(i)})\Big). \quad (27)$$

Since the covariance is fixed for this projection, the minimizer matches the first moment of the posterior:

$$\widehat{\mathbf{U}}^{(i-1)} = \mathbb{E}_{p(\mathbf{U}^{(i-1)}|\mathbf{U}^{(i)},\mathcal{C})}\left[\mathbf{U}^{(i-1)}\right]. \quad (28)$$

Substituting the posterior (25) into (28) and letting $\bar{\mathbf{U}}^{(i-1)}$ denote the integration variable, the posterior mean can be written as a ratio of expectations over the diffusion prior $\mathcal{N}(\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)})$:

$$\widehat{\mathbf{U}}^{(i-1)} = \frac{\mathbb{E}_{\mathcal{N}(\boldsymbol{\mu}^{(i)},\boldsymbol{\Sigma}^{(i)})}\left[\bar{\mathbf{U}}^{(i-1)} \exp\big(-J(\bar{\mathbf{U}}^{(i-1)})/\lambda\big)\right]}{\mathbb{E}_{\mathcal{N}(\boldsymbol{\mu}^{(i)},\boldsymbol{\Sigma}^{(i)})}\left[\exp\big(-J(\bar{\mathbf{U}}^{(i-1)})/\lambda\big)\right]}. \quad (29)$$

Thus, the constraint-conditioned posterior is approximated as

$$\widetilde{p}(\mathbf{U}^{(i-1)} \mid \mathbf{U}^{(i)},\mathcal{C}) = \mathcal{N}\Big(\mathbf{U}^{(i-1)}; \widehat{\mathbf{U}}^{(i-1)}, \boldsymbol{\Sigma}^{(i)}\Big), \quad (30)$$

from which the next reverse iterate is sampled. Equation (29) has the same algebraic form as the MPPI moment update in (12). GRACE therefore applies an MPPI-style posterior mean correction around the diffusion prior mean within each reverse step, instead of applying the guidance as an outer loop around the diffusion model.

In practice, we estimate (29) using the same MPPI estimator as in (14)–(15), with the sampling center replaced by $\boldsymbol{\mu}^{(i)}$. Candidate reverse samples are drawn as ${}^k\bar{\mathbf{U}}^{(i-1)} = \boldsymbol{\mu}^{(i)} + {}^k\delta\mathbf{U}^{(i)}$, where ${}^k\delta\mathbf{U}^{(i)} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}^{(i)})$, and are weighted by their rollout costs $J({}^k\bar{\mathbf{U}}^{(i-1)})$. GRACE then samples $\mathbf{U}^{(i-1)} \sim \mathcal{N}(\widehat{\mathbf{U}}^{(i-1)}, \boldsymbol{\Sigma}^{(i)})$.

Algorithm 1 GRACE inference procedure for gradient-free diffusion policy guidance.

Require: denoiser $\boldsymbol{\epsilon}_\theta$; state $\mathbf{x}_0$; cost J; reverse covariances $\{\boldsymbol{\Sigma}^{(i)}\}_{i=1}^N$; guidance covariances $\{\boldsymbol{\Sigma}_g^{(i)}\}_{i=1}^N$; samples K; temperature λ; execution horizon H_{exec}

Sample $\mathbf{U}^{(N)} \sim \mathcal{N}(0, \mathbf{I})$
for $i = N, N-1, \ldots, 1$ **do**
  Compute prior mean $\boldsymbol{\mu}^{(i)} = \boldsymbol{\mu}_\theta(\mathbf{U}^{(i)}, i)$ from (5)
  for $k = 1, \ldots, K$ **do**
    Sample ${}^k\delta\mathbf{U}^{(i)} \sim \mathcal{N}(0, \boldsymbol{\Sigma}_g^{(i)})$
    Form ${}^k\bar{\mathbf{U}}^{(i-1)} = \boldsymbol{\mu}^{(i)} + {}^k\delta\mathbf{U}^{(i)}$
    Roll out and evaluate $\widetilde{J}({}^k\bar{\mathbf{U}}^{(i-1)})$
  end for
  ${}^k w^{(i)} \leftarrow \dfrac{\exp\big(-\widetilde{J}({}^k\bar{\mathbf{U}}^{(i-1)})/\lambda\big)}{\sum_{l=1}^K \exp\big(-\widetilde{J}({}^l\bar{\mathbf{U}}^{(i-1)})/\lambda\big)}$
  $\hat{\mathbf{U}}^{(i-1)} \leftarrow \boldsymbol{\mu}^{(i)} + \sum_k {}^k w^{(i)} \cdot {}^k\delta\mathbf{U}^{(i)}$
  Sample $\mathbf{U}^{(i-1)} \sim \mathcal{N}\big(\widehat{\mathbf{U}}^{(i-1)}, \boldsymbol{\Sigma}^{(i)}\big)$
end for
Execute first H_{exec} actions of $\mathbf{U}^{(0)}$; replan.

Guidance covariance. The derivation above assumes the matched-covariance case, where the rollout sampling covariance equals the diffusion reverse covariance $\boldsymbol{\Sigma}^{(i)}$. Reusing it directly is problematic, since the schedule drives $\boldsymbol{\Sigma}^{(i)} \to \mathbf{0}$ as $i \to 1$, so the rollout perturbations vanish and the guidance can no longer explore as the action sequence is finalized. We therefore draw the rollouts from a separate guidance covariance $\boldsymbol{\Sigma}_g^{(i)}$ that sets the exploration scale independently of the denoising schedule. The sampling distribution $\mathcal{N}(\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}_g^{(i)})$ then no longer matches the measure in (29), and importance sampling reweights each rollout by the Gaussian ratio between the target and proposal densities,

$$\frac{\mathcal{N}(\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)})}{\mathcal{N}(\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}_g^{(i)})} \propto \exp\Big(-\tfrac{1}{2}\, \boldsymbol{\delta}^\top \big((\boldsymbol{\Sigma}^{(i)})^{-1} - (\boldsymbol{\Sigma}_g^{(i)})^{-1}\big)\, \boldsymbol{\delta}\Big), \quad (31)$$

where $\boldsymbol{\delta} := \bar{\mathbf{U}}^{(i-1)} - \boldsymbol{\mu}^{(i)}$. This absorbs into the Boltzmann factor as an augmented rollout cost

$$\widetilde{J}(\bar{\mathbf{U}}^{(i-1)}) = J(\bar{\mathbf{U}}^{(i-1)}) + \tfrac{\lambda}{2}\, \boldsymbol{\delta}^\top \big((\boldsymbol{\Sigma}^{(i)})^{-1} - (\boldsymbol{\Sigma}_g^{(i)})^{-1}\big)\, \boldsymbol{\delta}. \quad (32)$$

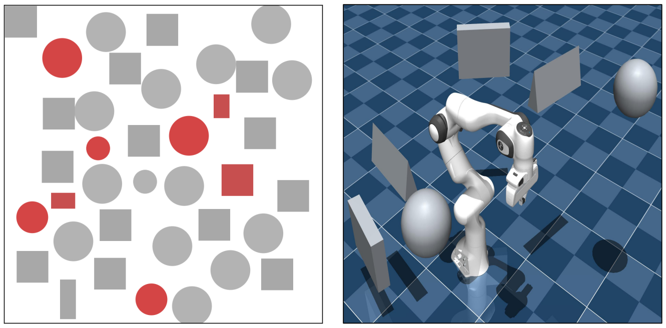
Replacing J with $\widetilde{J}$ in (29) yields an exact estimate of the posterior mean under the proposal $\boldsymbol{\Sigma}_g^{(i)}$.

Applying this transition recursively over all reverse steps yields the GRACE reverse process, summarized in Algorithm 1.

D. Connection to Gradient Guidance

Gradient-based diffusion guidance, as used in Motion Planning Diffusion (MPD) [15], approximates the same constraint-conditioned reverse posterior in (25) by locally linearizing the task log-likelihood around the diffusion prior mean $\boldsymbol{\mu}^{(i)}$. Specifically, it uses

$$\log p(\mathcal{C} \,|\, \mathbf{U}^{(i-1)}) \approx \log p(\mathcal{C} \,|\, \boldsymbol{\mu}^{(i)}) + (\mathbf{U}^{(i-1)} - \boldsymbol{\mu}^{(i)})^\top \mathbf{g}^{(i)}, \quad (33)$$



(a) (b)

Fig. 3. Simulation environments for the 2D and 3D planning tasks. (a) Planar point-mass navigation, where the red obstacles are absent during training and introduced only at inference time. (b) 7-DoF FR3 manipulator goal-reaching with randomly placed obstacles, all of which are introduced only at inference time.

where

$$\mathbf{g}^{(i)} := -\frac{1}{\lambda}\nabla_{\mathbf{U}} J(\mathbf{U})\big|_{\mathbf{U}=\boldsymbol{\mu}^{(i)}}. \tag{34}$$

The constant term $\log p(\mathcal{C} \mid \boldsymbol{\mu}^{(i)})$ is independent of $\mathbf{U}^{(i-1)}$ and is absorbed into the normalizer. Under this first-order approximation, the product of the linearized likelihood and the Gaussian diffusion prior remains Gaussian, giving the gradient-guided reverse update

$$\mathbf{U}^{(i-1)} \sim \mathcal{N}\Big(\boldsymbol{\mu}^{(i)} + \boldsymbol{\Sigma}^{(i)}\mathbf{g}^{(i)}, \boldsymbol{\Sigma}^{(i)}\Big). \tag{35}$$

Both methods target the same posterior (25), but approximate its mean differently. Gradient guidance linearizes the likelihood at $\boldsymbol{\mu}^{(i)}$, which requires differentiating both the cost and the dynamics, and reduces the correction to the single analytic direction $\boldsymbol{\Sigma}^{(i)}\mathbf{g}^{(i)}$ in (35). GRACE makes no such assumption, estimating the posterior mean from cost-weighted rollouts that only evaluate J on sampled candidates. It is therefore a more general guidance framework, extending the same posterior-mean estimate to the nondifferentiable rollout costs that arise from deployment-time constraints.

IV. EXPERIMENTS

In this section, we validate the effectiveness of GRACE on two simulated domains, a planar point-mass navigation task and a goal-reaching task on a 7-DoF Franka Research 3 (FR3) manipulator, as well as on a real-world experiment with the FR3 manipulator.

A. Experimental Setup

All experiments were conducted on a desktop computer with an Intel Core i7-13700F CPU at 2.1 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 4060 GPU, with all computation performed on the GPU. The diffusion prior is a goal-conditioned Diffusion Policy [11]. We use two interchangeable denoiser backbones, a FiLM-conditioned 1D temporal UNet **GRACE (UNet)** and a simpler FiLM-conditioned 1D residual CNN **GRACE (CNN)**, where the latter is included to examine whether a lighter model can still perform the tracking task well.

TABLE I
EXPERIMENTAL SETUP FOR THE TWO EVALUATION ENVIRONMENTS.

	Point-mass	FR3
Action	2D velocity	$\Delta\mathbf{q}$
Prior training data	10,000 (fixed)	20,000 (obs.-free)
Planning horizon	16	32
Executed per replan	8	16
Goal tolerance	0.05	0.03
Step budget	1,000	5,000
Obstacles at inference	fixed + added	6 random

Environments. Table I summarizes the setup of the two evaluation environments. The first is the planar point-mass navigation environment of Motion Planning Diffusion [15], shown in Fig. 3a, where the prior is trained on expert demonstrations generated using its open-source implementation on a map with a fixed set of obstacles. The start and goal are randomly generated, and additional obstacles absent during training are introduced only at inference time. A trial succeeds if the point mass reaches the goal within the goal tolerance, without collision or timeout. This environment poses two questions. The first is whether each method can find a feasible route to the goal in a cluttered map, including the obstacles unseen during training, evaluated over 30 random trials. The second is whether MPPI guidance preserves the multimodality of the prior, for the latter we collect 20 successful trajectories per method on a single fixed episode.

The second environment evaluates a 7-DoF Franka Research 3 (FR3) manipulator reaching goal joint configurations among 3D obstacles, shown in Fig. 3b, with the prior trained on obstacle-free feedback-tracking demonstrations. At inference time the obstacles together with the start and goal joint configurations are randomly generated, repeated over 30 random trials. A trial succeeds if the robot reaches the target within the joint tolerance without any safety violation or timeout. Here we ask whether GRACE can enforce the exact, nondifferentiable collision and joint-limit constraints while inheriting the feasibility of the prior.

Cost Formulation. Let $d_o(x_t)$ be the signed distance from state x_t to obstacle o, with penetration depth $p_o(x_t) = \max(0, -d_o(x_t))$. We define the obstacle cost over the horizon H as

$$C_{\text{obs}} = \sum_{t=1}^{H}\left(\mathbf{1}_{\text{obs}}(x_t) + \sum_{o\in\mathcal{O}}\left(\frac{p_o(x_t)}{\ell_o}\right)^2\right), \tag{36}$$

where $\mathbf{1}_{\text{obs}}$ is a binary overlap indicator, ℓ_o a per-obstacle normalization length, and the second term a soft-penetration penalty. The binary term is the exact collision indicator used for constraint checking, while the soft-penetration term is its continuous relaxation that supplies a gradient signal where the binary term provides none. For diffusion-guided methods the goal is supplied by the prior, so the total cost is

$$J = w_{\text{obs}}C_{\text{obs}} + w_{\text{prior}}C_{\text{prior}}, \tag{37}$$

where C_{prior} penalizes deviation from the prior, and the diffusion-free baselines replace C_{prior} with an explicit goal cost C_{goal}.

TABLE II
QUANTITATIVE COMPARISON OF PLANNING METHODS IN THE 2D POINT-MASS NAVIGATION TASK.

Method	Success [%]	Collision Fail	Path Length [m]
CEM	8/30 (26.7%)	0/30	1.247
MPPI	8/30 (26.7%)	0/30	1.241
DA-MPPI	8/30 (26.7%)	0/30	1.233
PO-DP	18/30 (60.0%)	12/30	2.657
GG-DP	18/30 (60.0%)	12/30	2.302
GRACE (CNN)	20/30 (66.7%)	0/30	3.681
GRACE (UNet)	25/30 (83.3%)	2/30	3.372

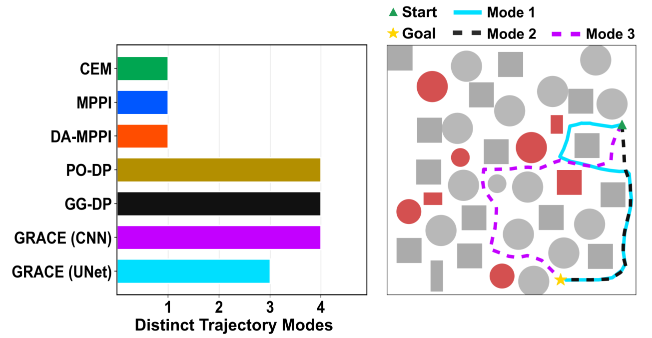
In the 2D environment all methods share the same cost in (36) over the exact geometry, so that the comparison isolates the difference between sampling- and gradient-based optimization under identical conditions. In the 3D environment the gradient-guided methods instead operate on a smoothed approximate geometry, whereas the sampling-based methods, which require no gradient, evaluate the exact geometry through the binary term alone. The 3D setting further adds self-collision, floor contact, and joint-limit constraints, aggregated as $C_{\text{safe}} = w_{\text{obs}}C_{\text{obs}} + w_{\text{self}}C_{\text{self}} + w_{\text{floor}}C_{\text{floor}} + w_{\text{jl}}C_{\text{jl}}$, which replaces $w_{\text{obs}}C_{\text{obs}}$ in (37).

Baselines. We compare against two families. The first consists of the diffusion-free sampling-based planners, namely **MPPI** [21], [23], **CEM** [24], and **DA-MPPI**, a Diffusion-Annealed MPPI (DA-MPPI) variant inspired by DIAL-MPC [25] that anneals the sampling variance across updates. These planners lack a learned prior and are pulled toward the goal by an explicit goal cost. For these planners and for the MPPI guidance in GRACE, we apply the same control perturbation across the time horizon for each sample, following the single-instance sampling scheme of [26], which improves sample efficiency and yields smoother rollouts. The second family is the gradient-guided diffusion policies, which share GRACE's prior but replace sampling-based guidance with gradient guidance over the differentiable surrogate cost. **Gradient-Guided Diffusion Policy (GG-DP)** applies in-loop refinement during denoising following MPD [15], and **Post-Optimized Diffusion Policy (PO-DP)** applies it once as a post-optimization pass after denoising.

B. Results

2D point-mass navigation. Table II summarizes the planar results. The diffusion-free planners, CEM, MPPI, and DA-MPPI, all reach 26.7% success with no collisions. They are conservative with respect to obstacles, but lacking any prior they are driven by the goal cost alone, so their uninformed proposal distributions cannot consistently discover feasible detours in the cluttered map under the sampling budget. They commit to a short, near-straight route toward the goal and steer into configurations from which no feasible route remains, a local minimum of the goal cost rather than a detour around the clutter. Their success-only path lengths are the shortest, around 1.24, because they succeed mainly on the easy configurations where such a short route already reaches the goal, so this should not be read as better planning quality.

The diffusion-based baselines exploit the learned prior, and both PO-DP and GG-DP reach 60.0% success, but they incur $12/30$ collision failures. This follows from the local nature of gradient guidance. When a predicted trajectory intersects an obstacle, the collision gradient pushes the colliding segment away from it, but in a cluttered map this locally corrective direction can move the segment toward a neighboring obstacle or into the narrow gap between obstacles, so the refinement oscillates between competing costs or settles in a local minimum. Deterministic gradient refinement alone therefore does not reliably resolve the narrow-gap constraint interactions, even with the prior.



(a) (b)

Fig. 4. 2D point-mass navigation results. (a) Number of trajectory modes produced by each method, measuring the diversity of generated solutions. (b) Representative trajectories of the three modes produced by GRACE (UNet), one per mode.

GRACE optimizes the same cost by sampling rather than gradient descent, combining the learned prior with sampling-based guidance that does not commit to a single gradient direction. This yields the strongest results in the planar task, with GRACE (CNN) reaching 66.7% success at zero collisions and GRACE (UNet) the best 83.3% at only $2/30$ collisions, surpassing both the diffusion-free and the gradient-guided planners. Its success-only paths are the longest, 3.681 for GRACE (CNN) and 3.372 for GRACE (UNet), because GRACE detours around clutter to solve the harder configurations that the other methods fail outright, so the added length reflects broader coverage rather than inefficiency.

We also assess whether the guidance preserves the multimodality of the prior. Fig. 4a reports the number of distinct trajectory modes per method. The diffusion-free planners show limited diversity, concentrating around a short route, while the diffusion-based methods produce multiple detour patterns, and GRACE retains this multimodality while improving collision satisfaction. This shows that MPPI guidance biases the prior toward lower-cost, constraint-satisfying samples without collapsing it onto a single path. Fig. 4b illustrates this for GRACE (UNet), showing a representative trajectory for each of its recovered modes.

3D goal-reaching. Table III reports the aggregate results, and Table IV breaks the failures down by cause. In contrast to the 2D navigation task, the diffusion-free sampling planners are competitive in the 3D manipulator domain, with CEM reaching 80.0% success and MPPI and DA-MPPI reaching 76.7%, compared with 36.7% for PO-DP and 50.0% for GG-DP. Their failures are concentrated in a single mode that matches how each updates its sampling distribution. MPPI

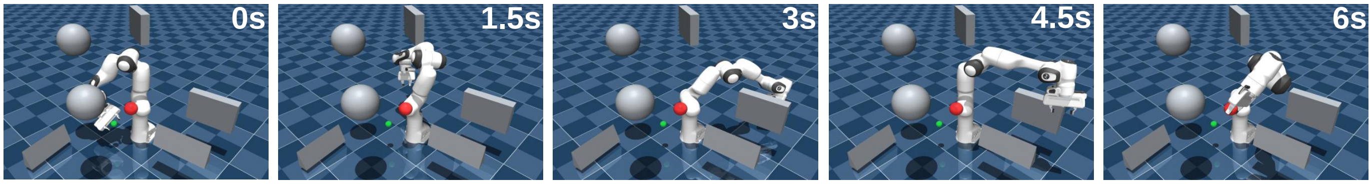


Fig. 5. Representative execution snapshots of GRACE in the 3D FR3 manipulator task. The manipulator reaches the target configuration while satisfying multiple safety constraints, including obstacle avoidance, self-collision avoidance, floor-contact avoidance, and joint-limit constraints.

TABLE III
QUANTITATIVE COMPARISON OF PLANNING METHODS IN THE 7-DOF FR3 MANIPULATOR GOAL-REACHING TASK.

Method	Success [%]	Computation Time [ms]	Guidance Time [ms]
CEM	24/30 (80.0%)	153.45	—
MPPI	23/30 (76.7%)	29.88	—
DA-MPPI	23/30 (76.7%)	154.22	—
PO-DP	11/30 (36.7%)	647.30	220.01
GG-DP	15/30 (50.0%)	671.59	244.72
GRACE (CNN)	27/30 (90.0%)	179.23	90.53
GRACE (UNet)	27/30 (90.0%)	497.64	85.88

TABLE IV
FAILURE MODES OF PLANNING METHODS IN THE 7-DOF FR3 MANIPULATOR GOAL-REACHING TASK.

Method	Obstacle	Self	Joint Limit	Floor	Max Steps
CEM	1	0	0	0	5
MPPI	2	0	5	0	0
DA-MPPI	6	0	0	0	1
PO-DP	6	2	6	2	3
GG-DP	5	2	3	1	4
GRACE (CNN)	3	0	0	0	0
GRACE (UNet)	3	0	0	0	0

keeps a fixed exploration variance and fails almost only on joint limits, consistent with its tendency to emit aggressive joint-space samples. DA-MPPI anneals this variance, and its failures are almost entirely obstacle collisions, consistent with too little late exploration to escape obstacle interactions. CEM refits to elite samples and rarely collides, with its failures being mostly timeouts from convergence to near-goal plateaus.

The gradient-guided diffusion policies perform worse despite using the same prior as GRACE, which shows that the gap is not explained by the prior but by how it is refined at inference time. Unlike the sampling planners, their failures are spread across every mode, and self-collision and floor contact occur only for these two methods. This pattern is a signature of gradient guidance. In the 7-DoF joint space all of the safety penalties enter the objective, but their gradients are merged into a single weighted update direction, so the update is pulled toward whichever term is locally largest while the others are left unsatisfied. Which constraint ends up violated then varies from trial to trial, scattering the failures across all modes rather than concentrating them in one. This local refinement is also computationally expensive, with PO-DP and GG-DP requiring 220.01 ms and 244.72 ms per guidance step, respectively.

GRACE achieves the best success rate, 90.0% for both backbones, with its only failures being 3 obstacle collisions per backbone and no violation of any other safety constraint. The modes that gradient guidance leaves unsatisfied therefore vanish entirely. Rather than committing to a single update direction, GRACE evaluates complete sampled rollouts around the prior and reweights them toward lower total cost, so the failures seen under gradient guidance are largely avoided. It thus preserves the kinematic feasibility of the obstacle-free prior while enforcing the safety costs introduced at inference time, and does so at a lower guidance cost of 90.53 ms and 85.88 ms per step than the gradient-guided policies. Fig. 5 shows representative execution snapshots of GRACE (UNet), where the manipulator reaches the target while avoiding the cluttered obstacles.

C. Real-World Experiments

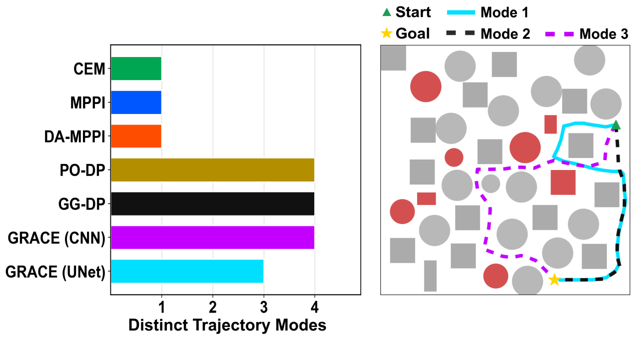
[Placeholder text for layout estimation. To be replaced with the final real-world setup and results.]

We further validate GRACE on a physical 7-DoF FR3 manipulator to confirm that the behavior observed in simulation transfers to a real robot. The same diffusion prior and guidance procedure are used without any retraining or fine-tuning, and the planner runs on the same desktop computer described above. The robot is commanded at the joint level through its real-time control interface, and the planned joint displacements are executed in the same chunked, receding-horizon manner as in simulation. Obstacles are placed in the workspace and their geometry is measured so that the corresponding collision cost is available to the guidance term at inference time. As in simulation, the prior is never exposed to obstacle information during training, and all collision avoidance is supplied online through the guidance term, so the only difference relative to the simulated setting is that the rollouts are evaluated against the physical workspace rather than a simulated one.

We design a set of cluttered goal-reaching scenarios in which the manipulator must move from a start configuration to a target configuration while avoiding the obstacles placed in the workspace. The scenarios are chosen to span a range of difficulties, from layouts with a single blocking obstacle to denser configurations that require the manipulator to find a detour around multiple obstacles. For each scenario we repeat the trial several times to account for the stochasticity of the sampling-based guidance, and we report the success rate together with the observed failure modes, using the same success and failure criteria as in the simulated FR3 experiments. We compare GRACE against the strongest baselines from the simulation study under identical obstacle layouts, so that the comparison isolates the effect of the guidance mechanism on real hardware.

V. CONCLUSION

In this paper, we presented Gradient-free Robot Action generation via Combined diffusion–MPPI posterior Estimation (GRACE) for constraint-aware diffusion planning. The key



(a) (b)

Fig. 6. Real world experiment.

result is that deployment-time guidance for diffusion policies does not have to rely on differentiable costs or backpropagated gradient updates. By using MPPI as a rollout-based posterior approximation, GRACE can refine a learned action prior with safety costs that are binary, discontinuous, or available only through forward evaluation. The simulation and real-world experiments show that this sampling-based guidance improves success and safety over gradient-guided diffusion baselines, especially when multiple constraints such as obstacle avoidance, self-collision, floor contact, and joint limits must be satisfied together. These results indicate that combining learned motion priors with gradient-free online cost evaluation is a practical direction for deploying diffusion policies in cluttered and constraint-rich robot environments.

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